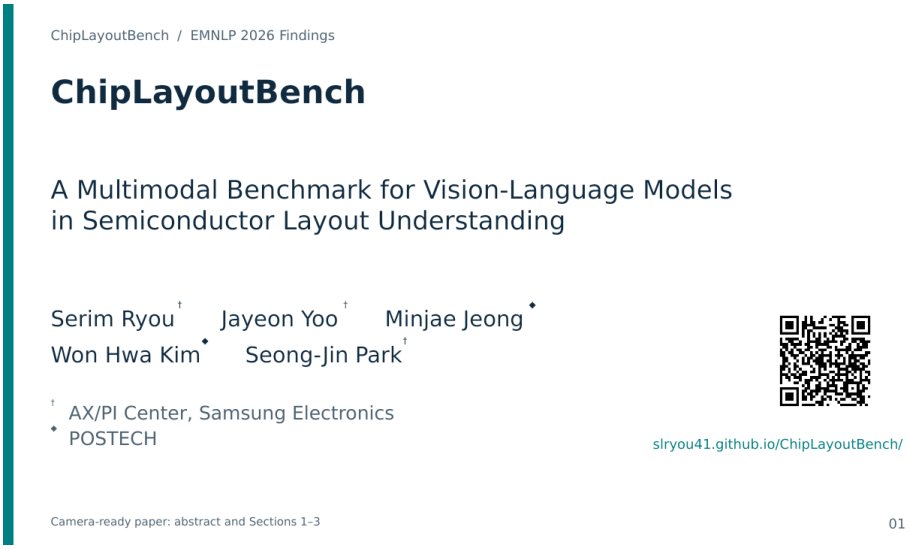
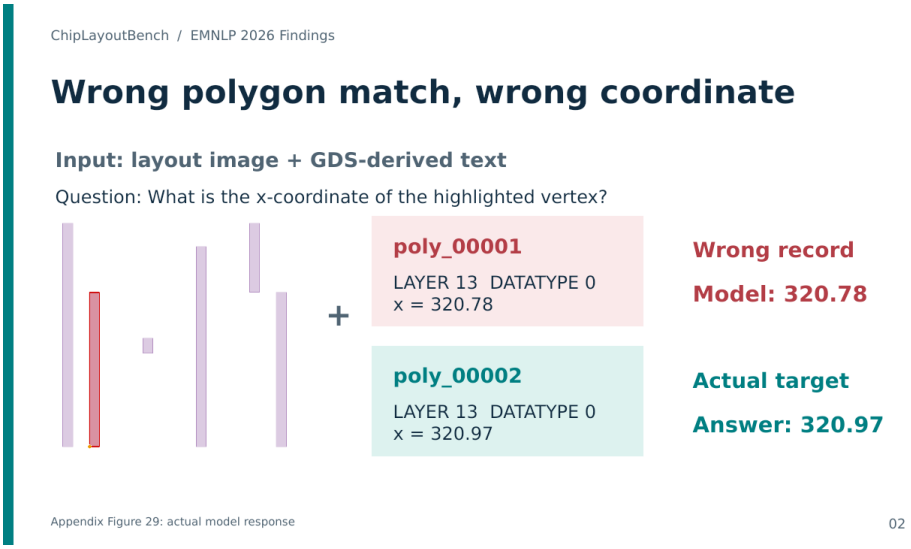
ChipLayoutBench



Approx. 42 seconds at a relaxed pace. Read the text below.

Hi, I’m Jayeon Yoo from Samsung Electronics. Today I’ll present ChipLayoutBench, a multimodal benchmark for semiconductor layout understanding. A chip layout has two complementary forms: a visual arrangement of shapes, and structured records containing exact geometry. We ask whether vision language models can connect the two. For example, when a question points to a polygon in an image, can the model find that same polygon in the supplied GDS text and use its coordinates correctly? We will start with an actual failure, explain the benchmark, and then compare performance across input representations.

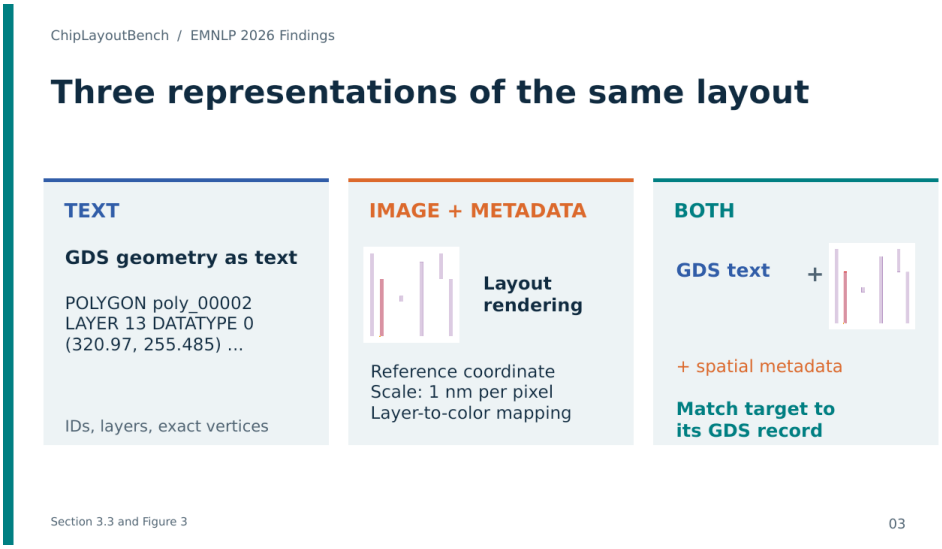
Wrong polygon match, wrong coordinate



Approx. 39 seconds at a relaxed pace. Read the text below.

Both the layout image and the GDS derived text are provided in this example. The question asks for the x coordinate of the highlighted vertex. The target belongs to polygon two, whose coordinate is 320.97. The model instead returns 320.78, from polygon one. Both numbers already exist in the input text. The model picks a coordinate directly from the supplied text, but fails to connect the highlighted target to the correct polygon record. The number is plausible, but it belongs to the wrong object.

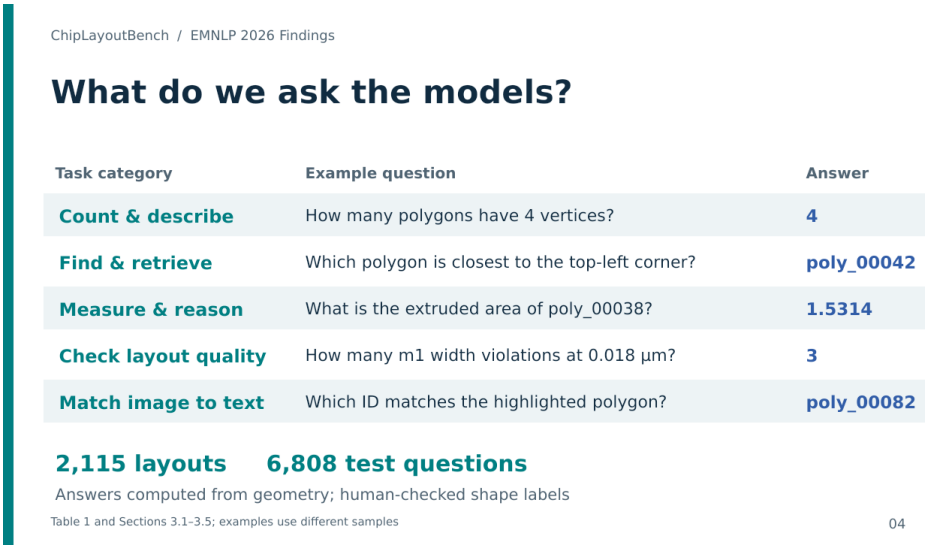
Three representations of the same layout



Approx. 43 seconds at a relaxed pace. Read the text below.

The benchmark compares three input settings. In text mode, the GDS geometry becomes a structured description. It contains polygon identifiers, layer and datatype attributes, and exact vertex coordinates. Image mode uses a rasterized semiconductor layout, accompanied by spatial metadata. This metadata supplies a reference coordinate, the physical length represented by each pixel, and a mapping from layers to colors. It does not supply the complete vertex lists. Both mode supplies the structured text together with the corresponding image and spatial reference. This comparison tests whether models can use visual context alongside exact geometric data.

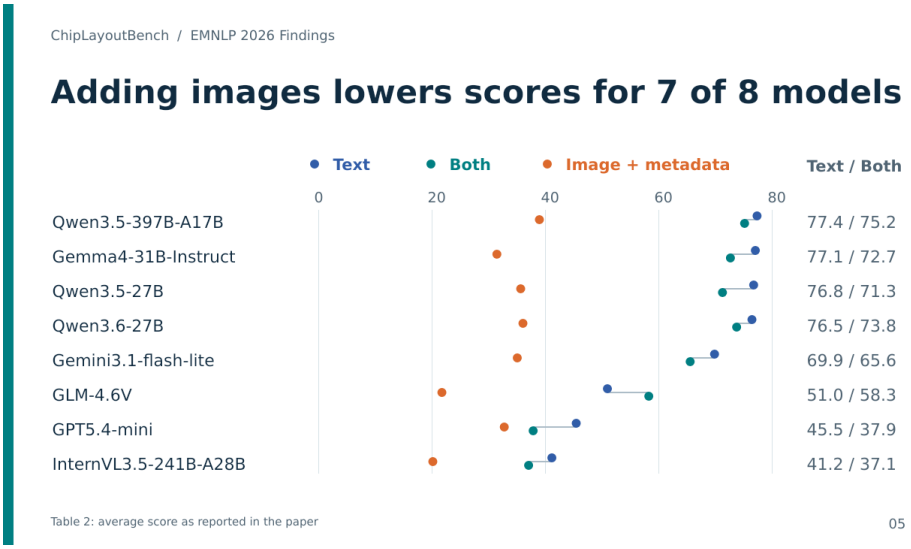
What do we ask the models?



Approx. 48 seconds at a relaxed pace. Read the text below.

We organize the benchmark into five task categories. Recognition asks questions such as how many polygons have four vertices. Retrieval asks for the polygon closest to a specified location. Geometric reasoning asks for an exact area, a distance, or a connectivity relationship. Layout tasks include counting design rule violations and detecting hotspots. The VLM specific category asks models to match highlighted image regions to polygon records and reason about visual targets. These examples illustrate different answer types: counts, polygon identifiers, and continuous quantities. The benchmark contains 2,115 layouts and 6,808 test questions. Most answers come from deterministic geometric computation, with human inspection for shape labels.

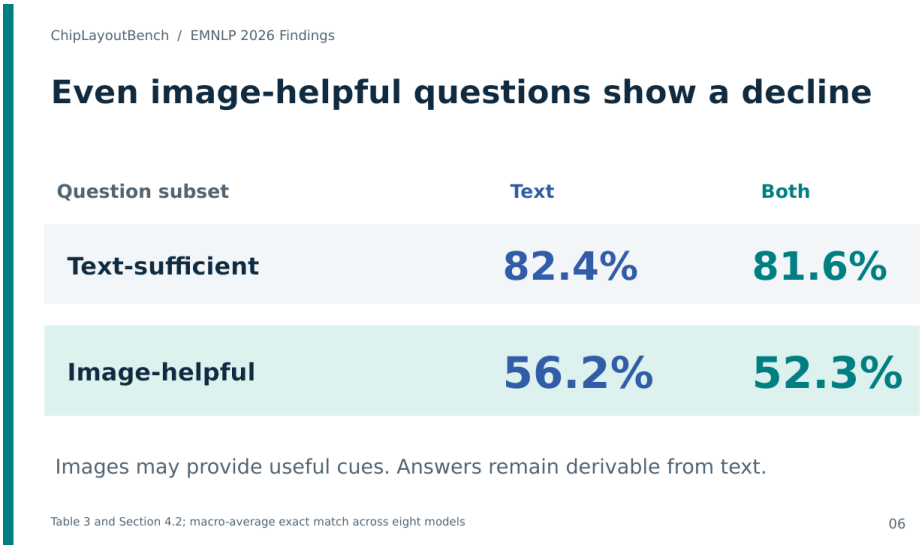
Adding images lowers scores for 7 of 8 models



Approx. 41 seconds at a relaxed pace. Read the text below.

Here are the average scores reported in Table 2 for all eight models. Blue represents text, teal represents both inputs, and orange represents the image with metadata. For seven models, the teal point is to the left of the blue point. GLM is the exception. The highest text score is 77.4, while the same model scores 75.2 with both inputs. Image plus metadata scores are lower than text for every model. The controlled comparison shows that adding visual context does not reliably improve exact layout reasoning in this benchmark.

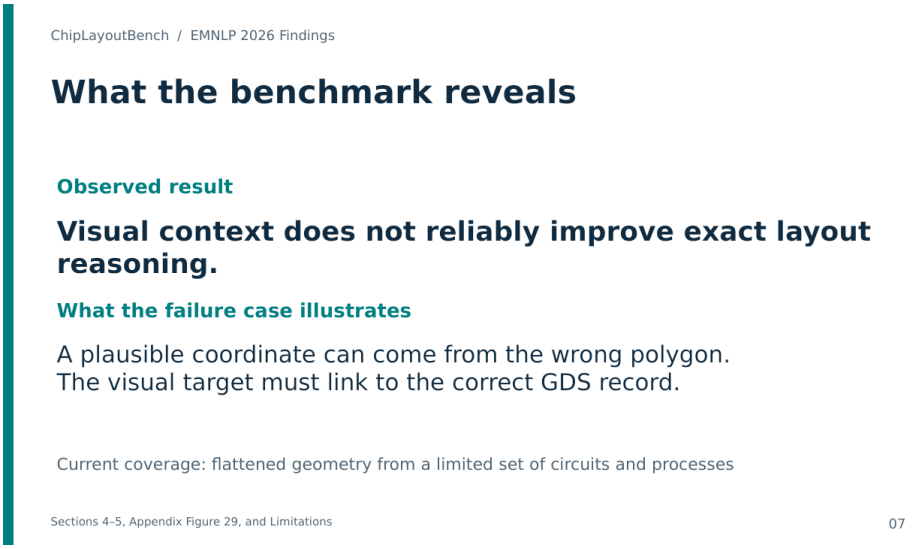
Even image-helpful questions show a decline



Approx. 42 seconds at a relaxed pace. Read the text below.

Many questions can already be answered from structured text. To examine this, the paper also separates text sufficient questions from image helpful questions. In the image helpful subset, visual cues such as shape or relative position may simplify the reasoning. Even here, the macro average exact match decreases from 56.2 percent with text to 52.3 percent with both inputs. These questions remain answerable from text, so this is not an image essential test. The result supports the broader observation that models do not consistently turn available visual information into better answers.

What the benchmark reveals



Approx. 48 seconds at a relaxed pace. Read the text below.

ChipLayoutBench evaluates how models connect semiconductor layout images to precise geometric data. Our experiments show that adding an image often reduces performance relative to structured text alone. The failure example gives one concrete illustration: a model can return a coordinate present in the input, while selecting the wrong polygon. This example illustrates the alignment challenge, although other errors can have different causes. A useful direction is to improve how models associate visual targets with structured elements before reasoning over their values. The benchmark currently uses flattened layouts and limited circuit and process coverage. Thank you for listening. The project page provides further information.